

Structured Covariance Estimation

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With Yonina Eldar (Weizmann Institute), Jerry Li (Microsoft Research),
and Cameron Musco (UMass Amherst).

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- Distribution $\mathcal{D}$ over d -dimensional vectors.
- $\mathbb{E}_{x \sim \mathcal{D}}[xx^T] = C$. $C_{j,k}$ is the covariance between x_j and x_k .

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How many samples $x^{(1)}, \dots, x^{(n)} \sim \mathcal{D}$ are required to learn C ?

Reasonable goal: Find $\tilde{C}$ with $\|C - \tilde{C}\|_2 \leq \epsilon \|C\|_2$.¹

¹Lots of other possible metrics.

Assuming $\mathcal{D}$ is Gaussian, subgaussian, subexponential:

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Known bound: $\Theta\left(\frac{d}{\epsilon^2}\right)$ samples are necessary and sufficient.

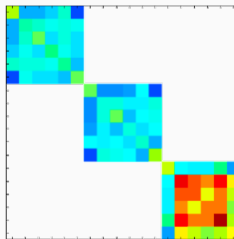
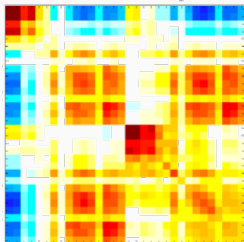
Estimator: Simple sample covariance.

$$\tilde{C} = \sum_{i=1}^n x^{(i)} x^{(i)T}.$$

Analysis: Standard matrix concentration (e.g., Vershynin, 2019).

What is we know C has additional structure?

- Block structure.
- Low-rank, low-rank + diagonal.
- Diagonal, banded.
- Many other possibilities.



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- C is rank- k : $\Theta\left(\frac{k}{\epsilon^2}\right)$, sample covariance.
- C is diagonal: $\Theta\left(\frac{\log d}{\epsilon^2}\right)$, estimate variance $C_{i,i}$ of each index separately. Set $C_{i,j} = 0$.

Not much known for more complicated structures...

SPATIALLY STRUCTURED COVARIANCE

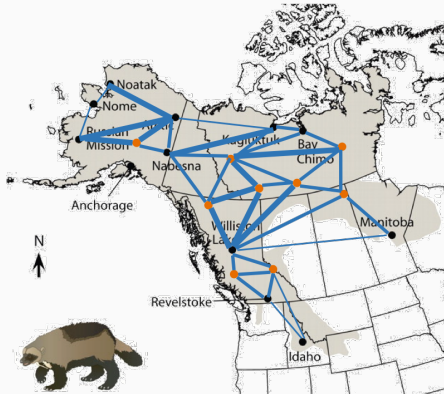
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Example: Graph structured genetic covariance in ecology.

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This work: Covariance matrix is Toeplitz.

$$T = \begin{bmatrix} a & b & c & d & e \\ b & a & b & c & d \\ c & b & a & b & c \\ d & c & b & a & b \\ e & d & c & b & a \end{bmatrix}$$

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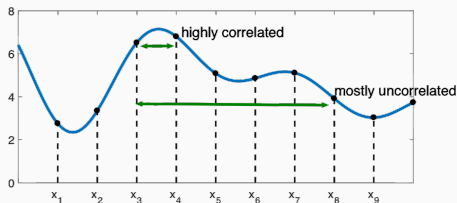
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²As for any covariance matrix, T must also be positive semidefinite.

TOEPLITZ COVARIANCE ESTIMATION

Arises when measurements taken on a spatial or temporal grid.

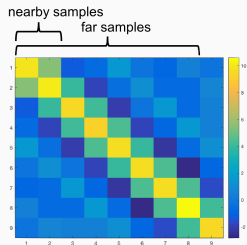
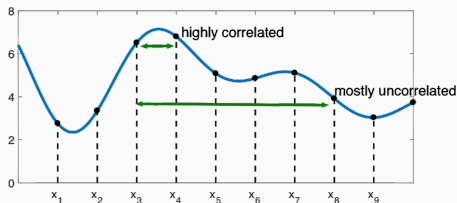
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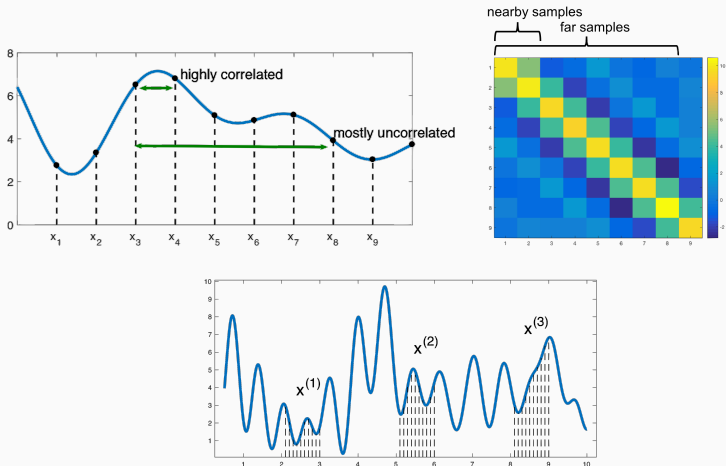
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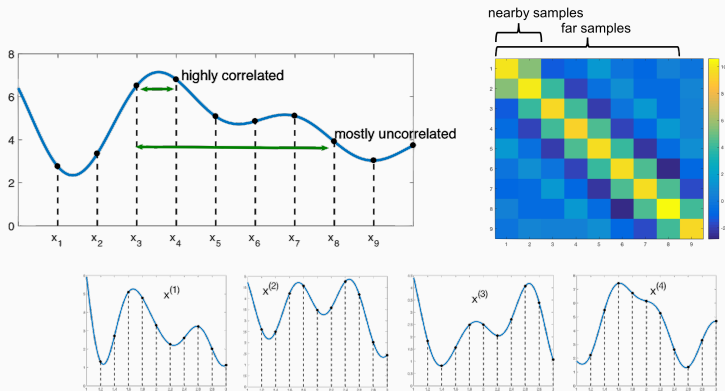
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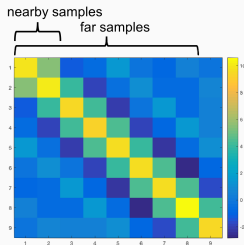
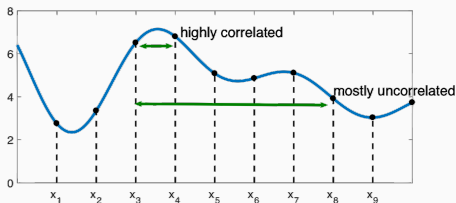
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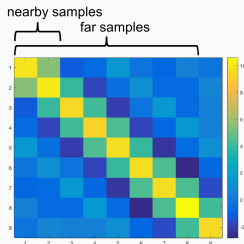
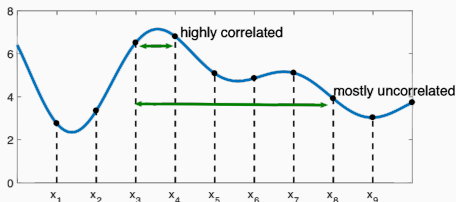


Applications in signal processing: spectrum sensing/cognitive radio, Doppler radar, direction-of-arrival estimation, prediction via Gaussian process regression, etc.

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Note: Shift-invariant kernel matrices in machine learning are Toeplitz covariance matrices when data points are on a grid.

SAMPLE COMPLEXITY

Goal: Minimize two types of sample complexity:

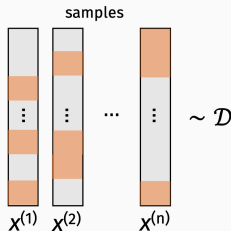
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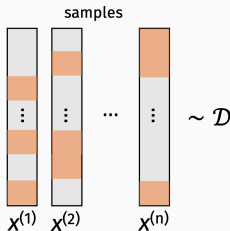
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In different applications, these complexities correspond to different costs. Typically there is a tradeoff.

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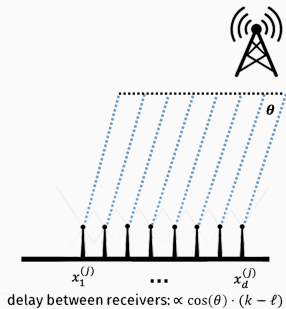
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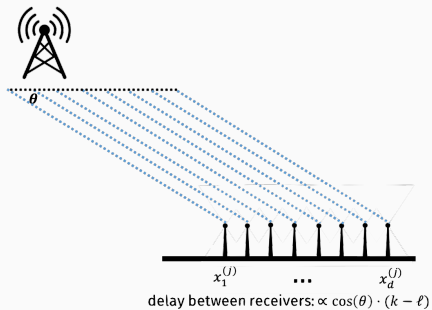
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- Seems to be interesting even beyond Toeplitz covariance matrices, but not well studied.

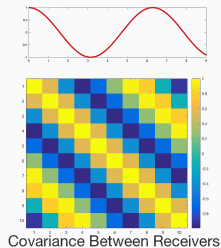
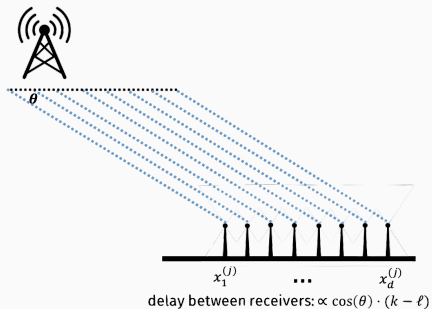
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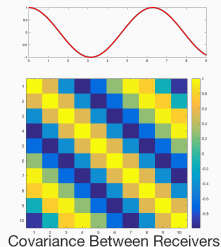
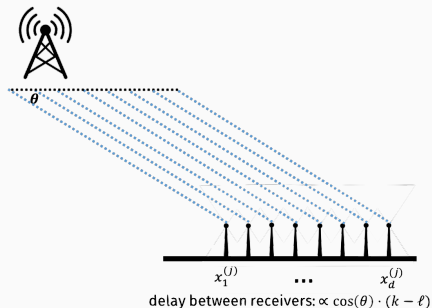


EXAMPLE: DIRECTION OF ARRIVAL ESTIMATION



Can back out direction of arrival θ from covariance structure.

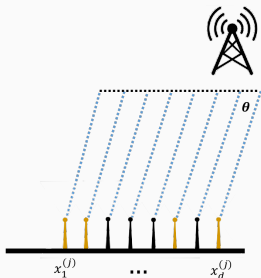
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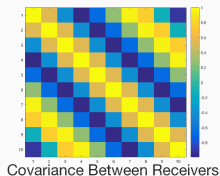
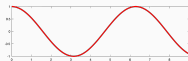
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delay between receivers $\propto \cos(\theta) \cdot (k - \ell)$



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- **Entry sample complexity:** Number of **active receivers**.

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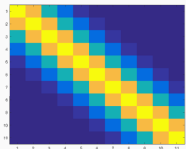
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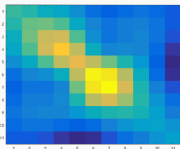
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- Show that sparse ruler methods give sublinear total sample complexity when T is low-rank (e.g., DOA with $k \ll d$ senders).
- Develop improved algorithms in the low-rank setting using techniques from matrix sketching, leverage score-based sampling, and sparse Fourier transforms. Resemble popular ‘subspace methods’ such as MUSIC and ESPRIT.

A FIRST RESULT

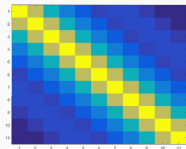
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True covariance T



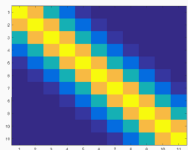
Empirical covariance $\hat{T}$



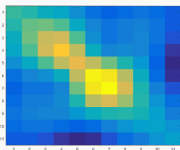
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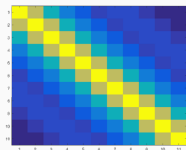
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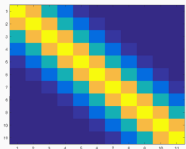


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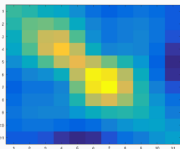
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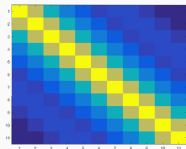
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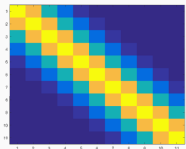


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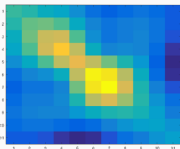
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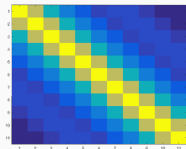
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Improves over $\tilde{O}(d^2 / \epsilon^2)$ for generic covariance matrices.

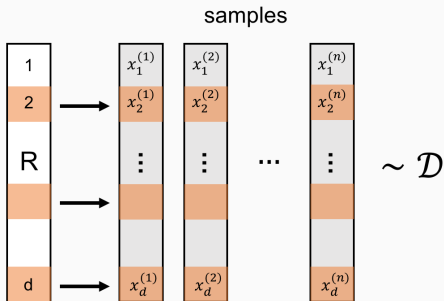
SUBSET BASED ESTIMATION

For today, consider algorithms that sample $x^{(1)}, \dots, x^{(n)} \sim \mathcal{D}$ and read a fixed subset of entries $R \subseteq [d]$ from each $x^{(j)}$. Approximate T using $x_R^{(1)}, \dots, x_R^{(n)} \in \mathbb{R}^{|R|}$.

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Entry sample complexity: $|R|$. Total sample complexity: $|R| \cdot n$.

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$$T_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{vs.} \quad T_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \textcolor{brown}{1} \\ 0 & 0 & 1 & 0 \\ 0 & \textcolor{brown}{1} & 0 & 1 \end{bmatrix}$$

To notice correlation between x_j and x_k must read both.

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How small can R be if T is Toeplitz? Can take advantage of redundancy.

$$T = \begin{bmatrix} a_0 & a_1 & a_2 & \cdots & a_{d-2} & a_{d-1} \\ a_1 & a_0 & a_1 & \cdots & \cdots & a_{d-2} \\ a_2 & a_1 & a_0 & \cdots & \cdots & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{d-2} & \cdots & \cdots & \cdots & \cdots & a_1 \\ a_{d-1} & a_{d-2} & \cdots & \cdots & a_1 & a_0 \end{bmatrix}$$

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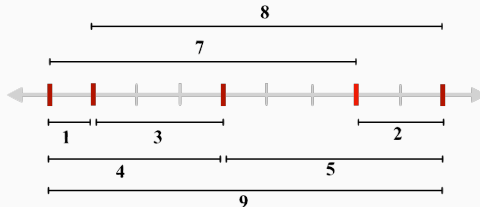
Will see that we can achieve $|R| = O(\sqrt{d})$.

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E.g., for $d = 10$, $R = \{1, 2, 5, 8, 10\}$ is a ruler.

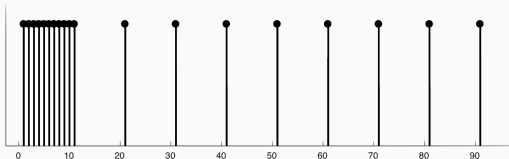


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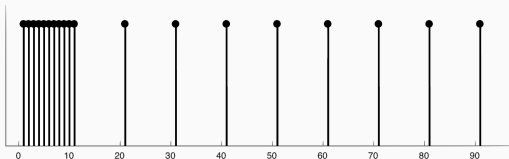


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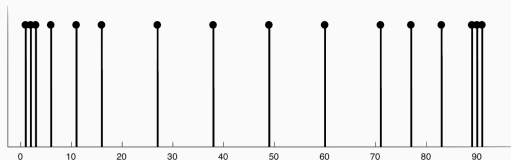
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$$T = \begin{bmatrix} a_0 & a_1 & a_2 & \cdots & a_{d-2} & a_{d-1} \\ a_1 & a_0 & a_1 & \cdots & \cdots & a_{d-2} \\ a_2 & a_1 & a_0 & \cdots & \cdots & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{d-2} & \cdots & \cdots & \cdots & \cdots & a_1 \\ a_{d-1} & a_{d-2} & \cdots & \cdots & a_1 & a_0 \end{bmatrix}$$

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- Get at least one independent sample of a_s from every $x_R^{(j)}$.
- With enough samples n from $\mathcal{D}$, will converge on an estimate of each a_s and so of the full matrix T .

How many vector samples do we need? What do we pay for the optimal entry sample complexity of sparse rulers?

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Theorem. For any ruler $R \subset [d]$, covariance estimation with R gives $\|\tilde{T} - T\|_2 \leq \varepsilon \|T\|_2$ with entry sample complexity $|R|$ and vector sample complexity $n = \tilde{O}\left(\frac{d}{\varepsilon^2}\right)$.

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- Vector sample complexity matches unstructured covariance estimation, but entry sample complexity is $O(\sqrt{d})$ instead of d .

Algorithm: For each $s \in \{0, \dots, d-1\}$ approximate a_s by average over the ruler R :

$$\tilde{a}_s = \frac{1}{n|R_s|} \sum_{j=1}^n \sum_{(k,\ell) \in R_s} x_k^{(j)} \cdot x_\ell^{(j)} \text{ where } R_s = \{k, \ell \in R : |k - \ell| = s\}.$$

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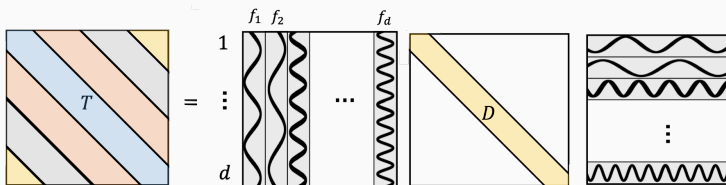
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- Leverage Fourier structure of Toeplitz matrices.

Vandermonde Decomposition: Any Toeplitz $T \in \mathbb{R}^{d \times d}$ can be written as $F_S D F_S$ where $F_S \in \mathbb{R}^{d \times d}$ is an ‘off-grid’ Fourier transform matrix with frequencies $f_1, \dots, f_d \in [0, 1]$ and D is a positive diagonal matrix.



$$F_S(j, k) = \exp(-2\pi i j \cdot f_k)$$

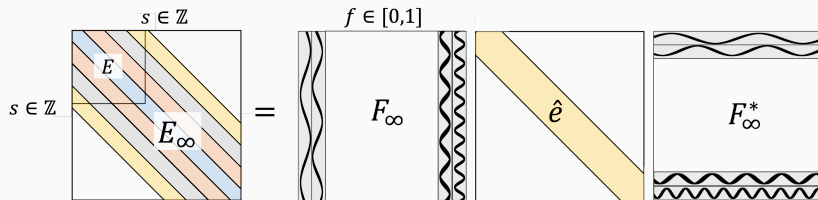
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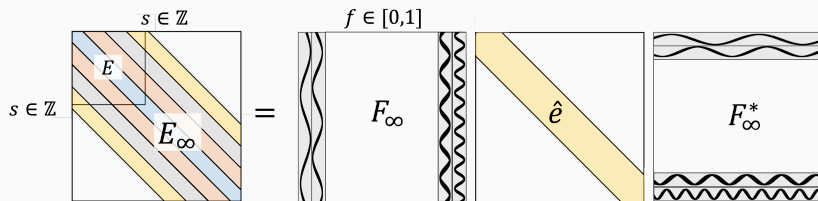
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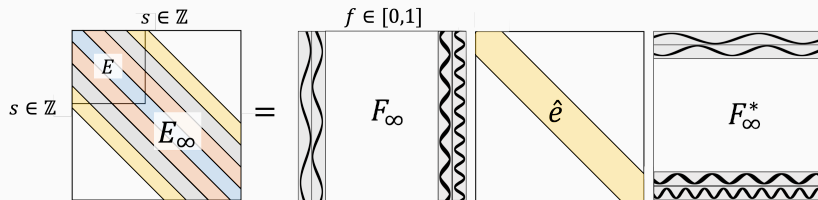


$$\|E\|_2 \leq \|E_\infty\|_2 = \max_{f \in [0,1]} \hat{e}(f)$$

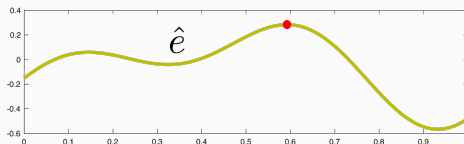
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$$\max_{f \in [0,1]} \hat{e}(f) \leq \epsilon \|T_R\|_2 \cdot \|M\|_F$$

T_R is the principal submatrix indexed by ruler R .

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- Let's us **interpolate between minimal entry sample complexity and minimal vector sample complexity.**

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Sparse-ish rulers give the following tradeoff for $\alpha \geq 1/2$:

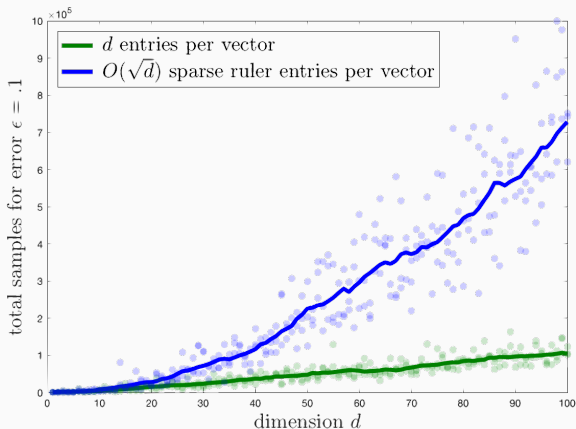
- Entry sample complexity: $O(d^\alpha)$.
- Vector sample complexity: $O(d^{2-2\alpha})$.

SPARSE RULER VS. FULL RULER

Total sample complexity is $O(\sqrt{d}) \cdot \tilde{O}(d) = \tilde{O}(d^{3/2})$ for sparse ruler vs. $d \cdot \tilde{O}(1) = \tilde{O}(d)$ for full sample estimation.

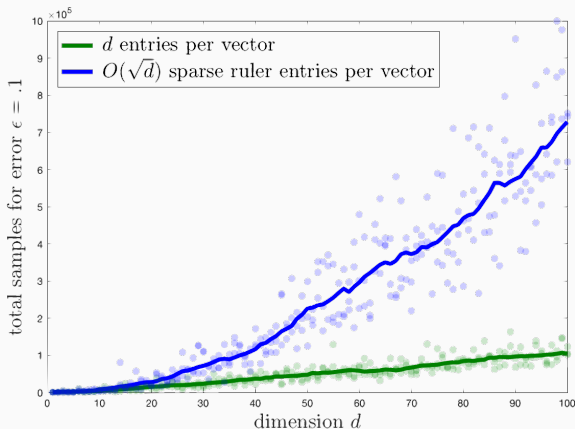
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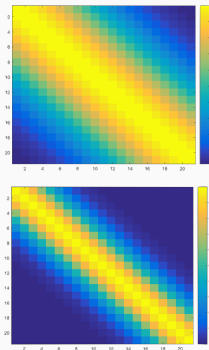
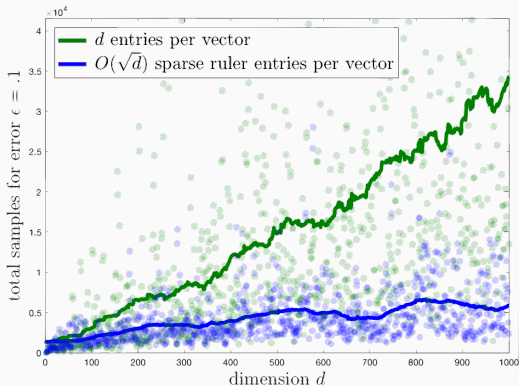
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- Prove bounds are tight when T is the identity.

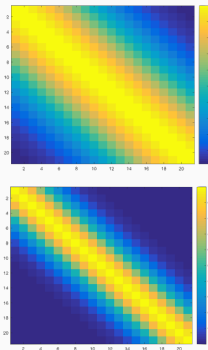
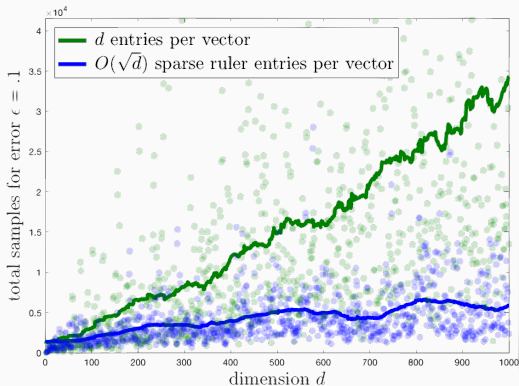
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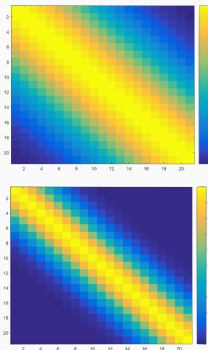
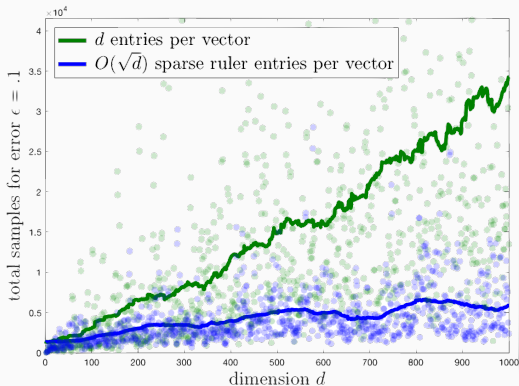
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- Total sample complexity is $\tilde{O}(\sqrt{d})$ for sparse rulers vs. $\tilde{O}(d)$ for full samples. Can we explain this?

Recall that we have with $n = \tilde{O}(1/\varepsilon^2)$ samples:

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SPARSE RULERS FOR LOW-RANK MATRICES

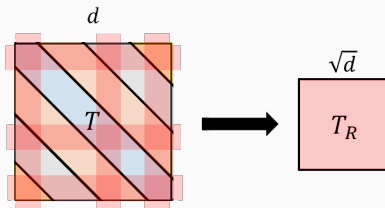
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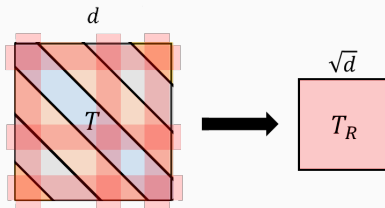
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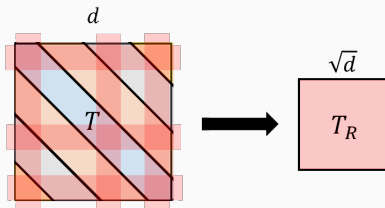


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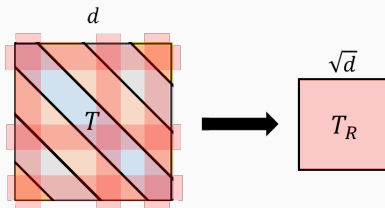


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- Low-rank matrices cannot look like the identity – have significant off diagonal mass [Musco, Musco, Woodruff ‘19].

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Recall that we have with $n = \tilde{O}(1/\varepsilon^2)$ samples:

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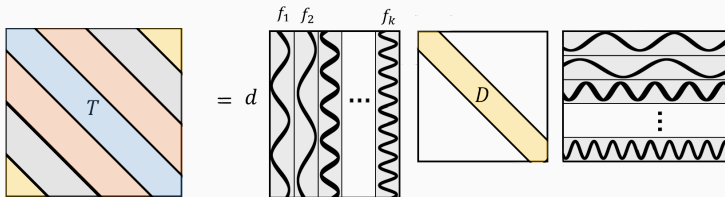


- If T is the identity, $\|T\|_2 = \|T_R\|_2 = 1$. But this is ‘very’ full-rank.
- Low-rank matrices cannot look like the identity – have significant off diagonal mass [Musco, Musco, Woodruff ‘19].
- **Upshot:** Show $\|T_R\|_2 \leq \frac{k}{\sqrt{d}} \|T\|_2$. Setting $\varepsilon' = \varepsilon/k$ obtain total sample complexity $\tilde{O}(\sqrt{d}k^2/\varepsilon^2)$.

Remainder of the talk: Sketch a new approach to low-rank Toeplitz covariance estimation using sparse Fourier transform methods.

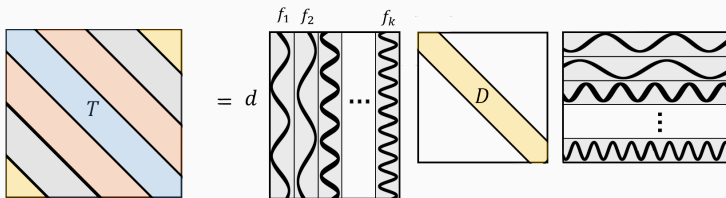
THE FOURIER PERSPECTIVE

Vandermonde Decomposition: Any rank- k Toeplitz $T \in \mathbb{R}^{d \times d}$ can be written as $F_S D F_S^T$ where $F_S \in \mathbb{R}^{d \times k}$ is an ‘off-grid’ Fourier transform matrix with frequencies $f_1, \dots, f_k$ and D is a positive diagonal matrix.



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- Any sample $x \sim \mathcal{N}(0, T)$ can be written as $T^{1/2}g = F_S D^{1/2}g$ for $g \sim \mathcal{N}(0, I)$. $\mathbb{E}[xx^T] = F_S D^{1/2} \mathbb{E}[gg^T] D^{1/2} F_S^T = T$.

$x \sim \mathcal{N}(0, T) = F_s D^{1/2} g$ is a Fourier sparse function.

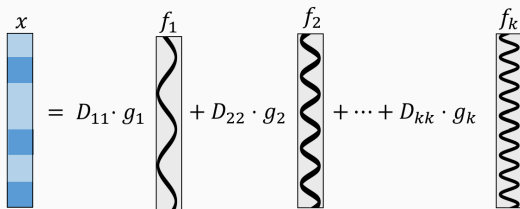
SAMPLE SECOVERY VIA SPARSE FOURIER FRANSFORM

$x \sim \mathcal{N}(0, T) = F_S D^{1/2} g$ is a **Fourier sparse function**.

$$x = D_{11} \cdot g_1 + D_{22} \cdot g_2 + \cdots + D_{kk} \cdot g_k$$

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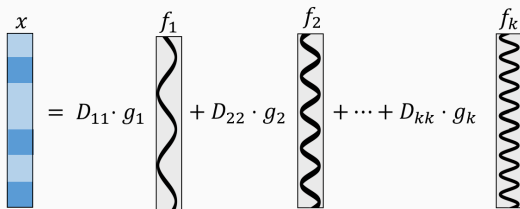
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- Take $n = \tilde{O}(1/\varepsilon^2)$ samples, recover each in full by reading $2k$ entries, and then apply our earlier result for full ruler $R = [d]$. Total sample complexity: $\tilde{O}(k/\varepsilon^2)$.

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- Well studied in TCS, but mostly in the case when $f_1, \dots, f_k$ are 'on grid' frequencies.

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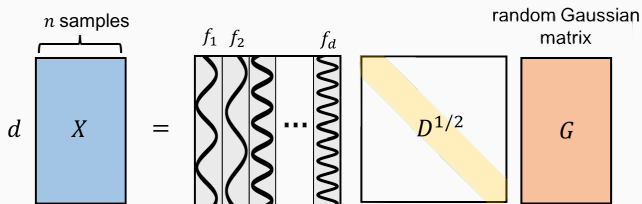
Theorem: Any $A \in \mathbb{R}^{n \times d}$, contains a subset of $O(k/\varepsilon)$ columns, C such that:

$$\|A - P_C \cdot A\|_F^2 \leq (1 + \varepsilon) \min_{M, \text{rank}(M)=k} \|A - M\|_F^2.$$

$x^{(1)}, \dots, x^{(n)} \sim \mathcal{N}(0, T)$ can be written as $X = F_S D^{1/2} G$ where columns of G are distributed as $\mathcal{N}(0, I)$.

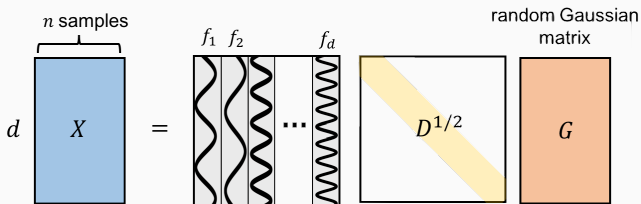
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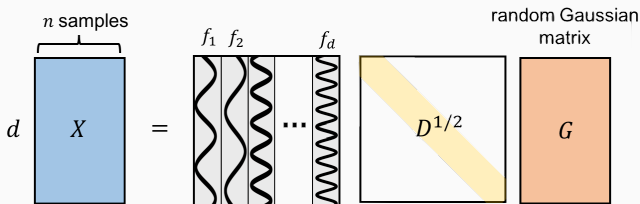
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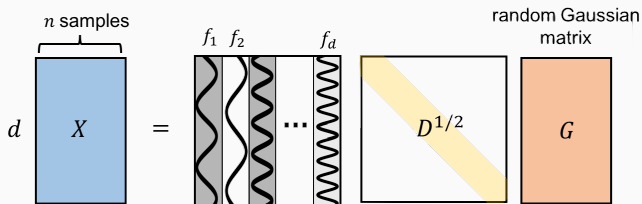
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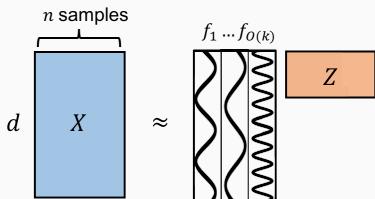
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- Find frequencies via brute force search over a net.
- At each step of the search, for a given F_M , we must find Z that reconstructs X as well as possible using these frequencies. How do we do this without reading all of X ?

APPROXIMATE FREQUENCY REGRESSION

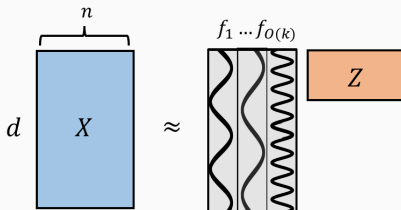
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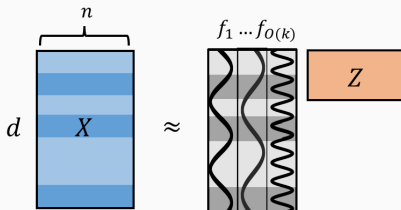
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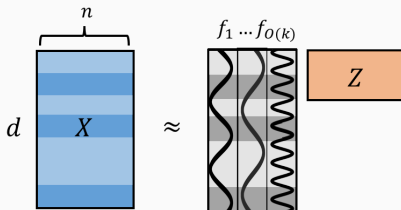


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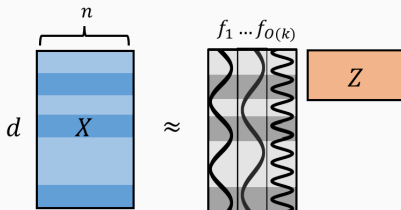


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FOURIER LEVERAGE SCORES

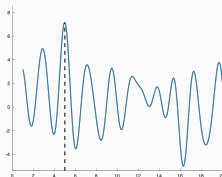
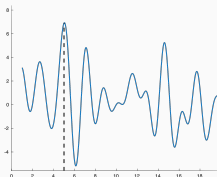
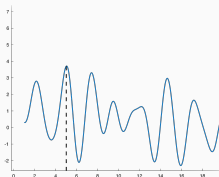
Leverage scores measure how large a function in the column span of F_M can be at index i (i.e., how important that index may be in the regression.)

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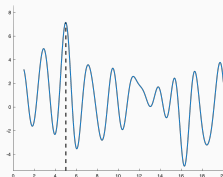
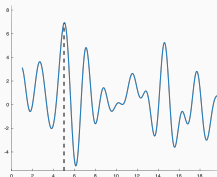
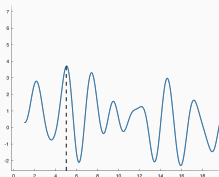
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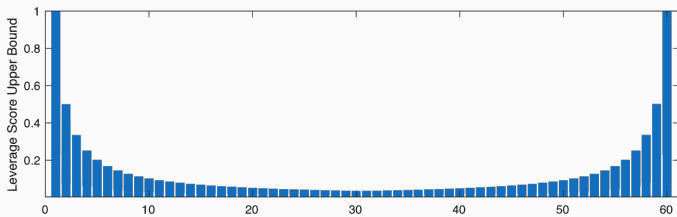
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- Using that $F_M y$ is a Fourier sparse function we can bound this quantity a priori, without any dependence on F_M .

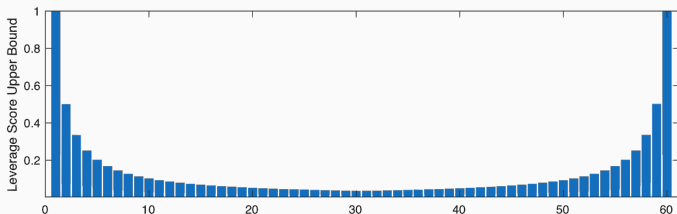
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Since this distribution is universal, can sample one set of entries by these leverages scores, and find $X \approx F_M \cdot Z$ with high probability for any set of frequencies $f_1, \dots, f_m$ in net.

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Sample Complexity: Vector, entry, total sample complexity:
 $\tilde{O}(\text{poly}(k/\epsilon))$.

Bound: $\|T - \tilde{T}\|_2 \leq \epsilon \|T\|_2 + f(T - T_k)$

OPEN QUESTIONS AND FUTURE DIRECTIONS

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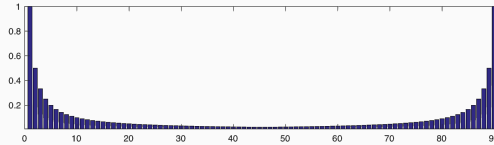
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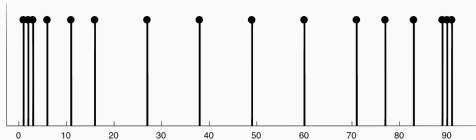
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 - Sample complexity bounds and tradeoffs for applications like direction-of-arrival estimation, Doppler imaging.

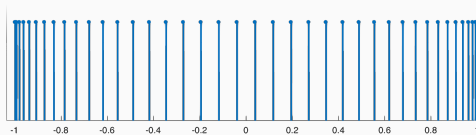
CONNECTIONS BETWEEN SAMPLING SCHEMES



Fourier Sparse Leverage Scores

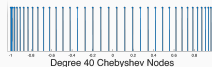
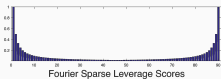


Optimal Sparse Ruler for $d=91$



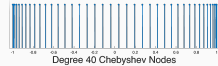
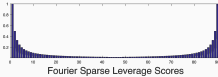
Degree 40 Chebyshev Nodes

CONNECTIONS BETWEEN SAMPLING SCHEMES



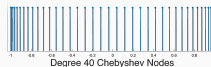
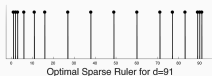
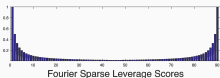
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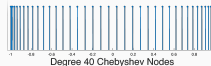
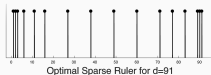
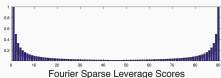
CONNECTIONS BETWEEN SAMPLING SCHEMES



- **Some Formal Connections:**

- Limiting density of Chebyshev nodes is the leverage score distribution for k degree polynomials.
- Sampling $\tilde{O}(\sqrt{d})$ indices via Fourier sparse leverage scores gives a sparse ruler with good probability.

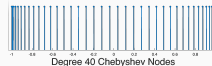
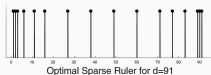
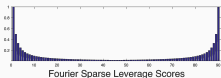
CONNECTIONS BETWEEN SAMPLING SCHEMES



- **Some Formal Connections:**

- Limiting density of Chebyshev nodes is the leverage score distribution for k degree polynomials.
- Sampling $\tilde{O}(\sqrt{d})$ indices via Fourier sparse leverage scores gives a sparse ruler with good probability.
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- Seem to have a lot more to understand.

THANKS! QUESTIONS?